

AI Impact on Jobs 2030 - Descriptive Analysis

Notebook 1 : Introduction

As artificial intelligence continues to transform the workplace, questions arise about which jobs are most at risk of automation. Some occupations may be highly exposed to AI-driven technologies, while others rely on uniquely human skills that are less easily automated.

This project draws on a dataset simulating the future of work in 2030, including 3,000 observations with information on salary, experience, education, AI exposure, and automation risk.

The primary question guiding this descriptive analysis is:

“Which types of jobs are most at risk of automation, and how do job characteristics are distributed across different levels of automation risk?”

Through this exploration, we aim to identify patterns and insights that highlight the potential impact of AI on the workforce, without making predictive or causal claims.

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Dataset: AI Impact on Jobs 2030 — Kaggle



AI Illustration: Man and robot working together in a high-tech office (Freepik)

Data Loading and Overview

First 10 rows of the AI Impact on Jobs 2030 dataset

Job_Title	Average_Salary	Years_Experience	Education_Level	AI_Exposure_Index	Tech_Growth_Factor	Automation_Probability_2030	Risk_C
Security Guard	45795	28	Master's	0.18	1.28	0.85	High
Research Scientist	133355	20	PhD	0.62	1.11	0.05	Low
Construction Worker	146216	2	High School	0.86	1.18	0.81	High
Software Engineer	136530	13	PhD	0.39	0.68	0.60	Medium
Financial Analyst	70397	22	High School	0.52	1.46	0.64	Medium
AI Engineer	92592	11	Master's	0.29	0.51	0.10	Low
Mechanic	107373	23	PhD	0.67	1.09	0.41	Medium
Teacher	53419	12	High School	0.20	1.40	0.17	Low
HR Specialist	139225	12	Master's	0.30	0.61	0.48	Medium
Customer Support	85016	2	High School	0.01	1.01	0.80	High

Description of Dataset Columns for AI Impact on Jobs 2030

Column	Description
Job_Title	Name of the occupation (e.g., Data Scientist, Truck Driver)
Average_Salary	Average annual salary for the job (USD)
Years_Experience	Average years of experience typically required
Education_Level	Highest common level of education required (High School, Bachelor's, Master's, PhD)
AI_Exposure_Index	How much the job interacts with AI tools (0 = no exposure, 1 = highly AI-dependent)
Tech_Growth_Factor	Multiplier representing how quickly technology is advancing in the field
Automation_Probability_2030	Estimated probability that the job will be automated by 2030
Risk_Category	Categorized automation risk: Low, Medium, or High
Skill_1	Randomized skill proficiency levels for one of 10 skill dimensions
Skill_2	Randomized skill proficiency levels for one of 10 skill dimensions
Skill_3	Randomized skill proficiency levels for one of 10 skill dimensions
Skill_4	Randomized skill proficiency levels for one of 10 skill dimensions
Skill_5	Randomized skill proficiency levels for one of 10 skill dimensions
Skill_6	Randomized skill proficiency levels for one of 10 skill dimensions
Skill_7	Randomized skill proficiency levels for one of 10 skill dimensions
Skill_8	Randomized skill proficiency levels for one of 10 skill dimensions
Skill_9	Randomized skill proficiency levels for one of 10 skill dimensions
Skill_10	Randomized skill proficiency levels for one of 10 skill dimensions

General Information about the Dataset

Metric	Value
Number of rows	3000
Number of columns	18
Duplicate rows	0

Variables Overview

	Column	Type	Missing	Missing_Percent
Job_Title	Job_Title	character	0	0

	Column	Type	Missing	Missing_Percent
Average_Salary	Average_Salary	numeric	0	0
Years_Experience	Years_Experience	numeric	0	0
Education_Level	Education_Level	character	0	0
AI_Exposure_Index	AI_Exposure_Index	numeric	0	0
Tech_Growth_Factor	Tech_Growth_Factor	numeric	0	0
Automation_Probability_2030	Automation_Probability_2030	numeric	0	0
Risk_Category	Risk_Category	character	0	0
Skill_1	Skill_1	numeric	0	0
Skill_2	Skill_2	numeric	0	0
Skill_3	Skill_3	numeric	0	0
Skill_4	Skill_4	numeric	0	0
Skill_5	Skill_5	numeric	0	0
Skill_6	Skill_6	numeric	0	0
Skill_7	Skill_7	numeric	0	0
Skill_8	Skill_8	numeric	0	0
Skill_9	Skill_9	numeric	0	0
Skill_10	Skill_10	numeric	0	0

Statistical Summary of Numeric Variables

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
Average_Salary	30030.00	58640.00	89318.000	89372.279	119086.50	149798.00
Years_Experience	0.00	7.00	15.000	14.678	22.00	29.00
AI_Exposure_Index	0.00	0.26	0.500	0.501	0.74	1.00
Tech_Growth_Factor	0.50	0.74	1.000	0.995	1.24	1.50
Automation_Probability_2030	0.05	0.31	0.500	0.502	0.70	0.95
Skill_1	0.00	0.24	0.505	0.497	0.74	1.00
Skill_2	0.00	0.25	0.500	0.497	0.74	1.00
Skill_3	0.00	0.25	0.500	0.499	0.75	1.00
Skill_4	0.00	0.26	0.510	0.504	0.75	1.00
Skill_5	0.00	0.24	0.490	0.490	0.73	1.00
Skill_6	0.00	0.26	0.500	0.500	0.74	1.00
Skill_7	0.00	0.25	0.490	0.499	0.75	1.00
Skill_8	0.00	0.25	0.500	0.503	0.75	1.00
Skill_9	0.00	0.26	0.500	0.501	0.74	1.00
Skill_10	0.00	0.25	0.490	0.494	0.74	1.00

Summary of Initial Findings

This initial descriptive exploration confirms that the AI Impact on Jobs 2030 dataset is well-structured and suitable for illustrating patterns related to automation risk. Covering 3,000 observations for 20 different job titles with no missing values and no duplicates, the dataset includes a balanced mix of economic, educational, and technological variables. Although the data are simulated, they provide a clear framework for exploring automation probabilities, AI exposure, and risk categories across different job types.

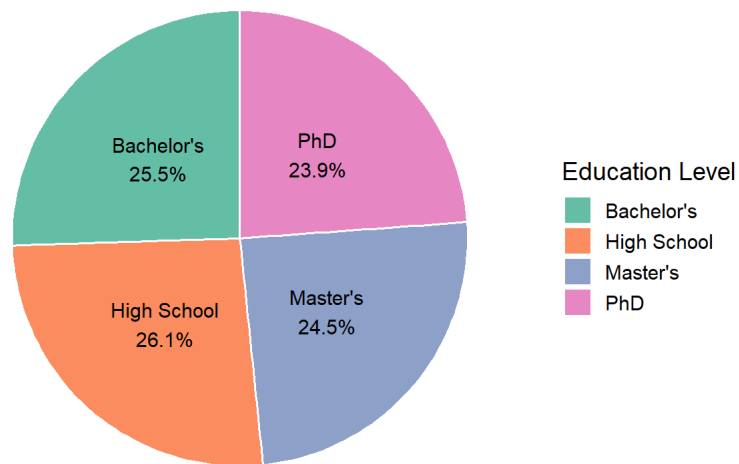
Preliminary statistics indicate substantial variation across jobs in terms of salary, experience, AI exposure, and estimated automation probability, with central tendencies suggesting moderate average exposure to AI and a wide dispersion of automation risk by 2030. The diversity observed across education levels, risk categories, and skill profiles highlights the heterogeneity of the future labor market and suggests that automation risk is unlikely to be uniform across occupations.

For the purpose of this project, the ten skill variables (Skill_1 to Skill_10) are excluded from the initial univariate analysis. These variables are randomized proficiency levels representing different skill dimensions (e.g., creativity, data analysis, robotics). Focusing on variables like Years_Experience, AI_Exposure_Index, Tech_Growth_Factor, and Automation_Probability_2030 allows for clear, interpretable visualizations that highlight meaningful patterns in automation risk.

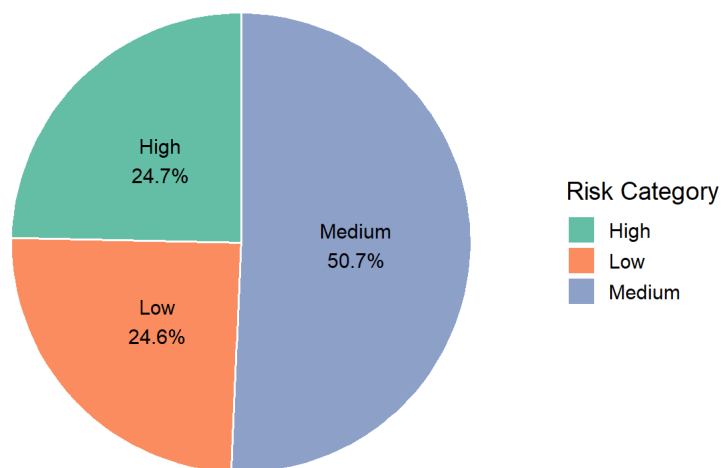
These initial findings provide a solid foundation for deeper descriptive analyses examining how job characteristics such as experience, education, and AI exposure relate to automation vulnerability, without making predictive or causal claims.

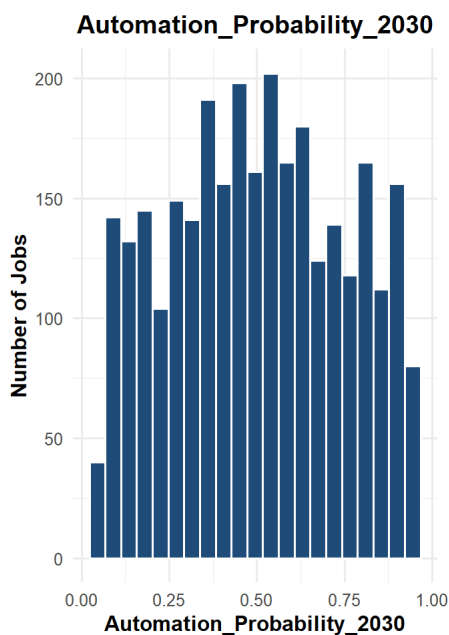
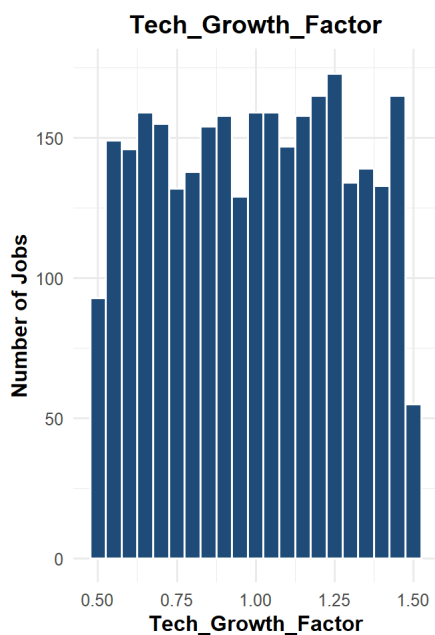
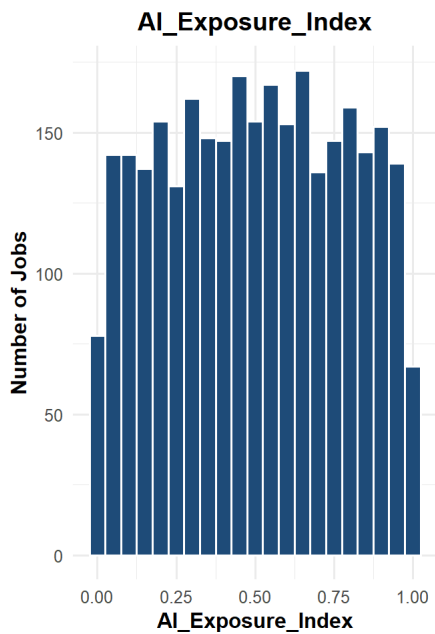
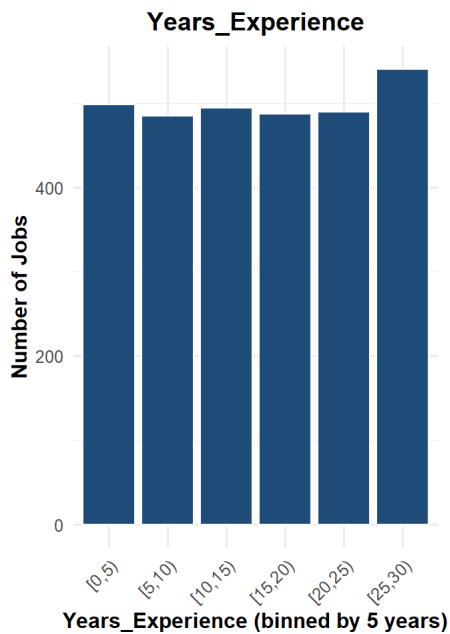
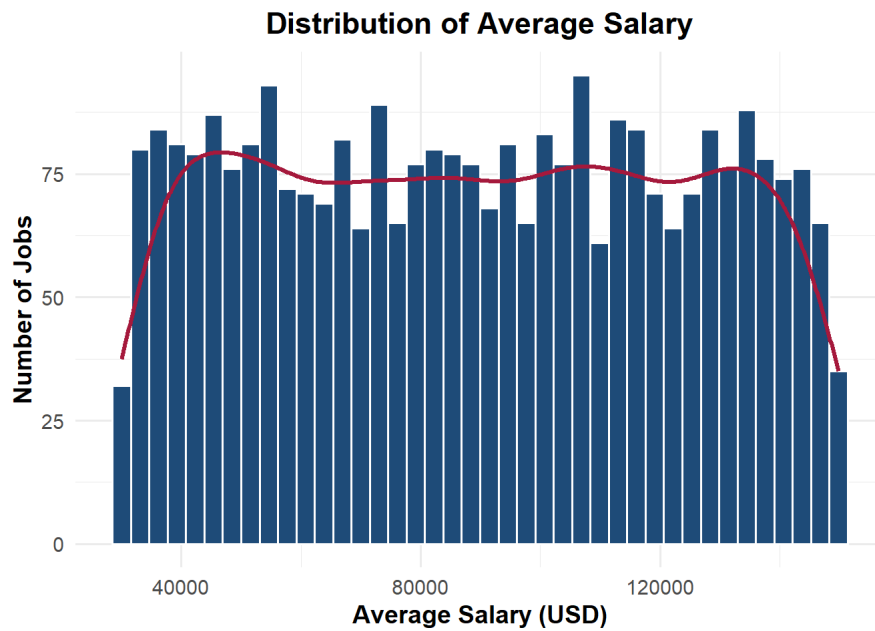
Univariate Analysis

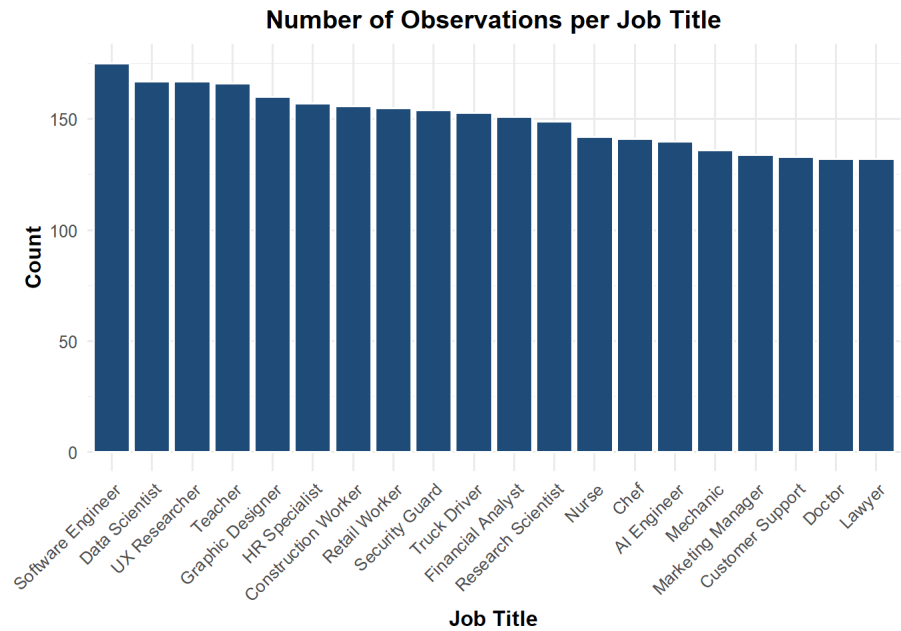
Distribution of Education Levels



Distribution of Automation Risk Categories



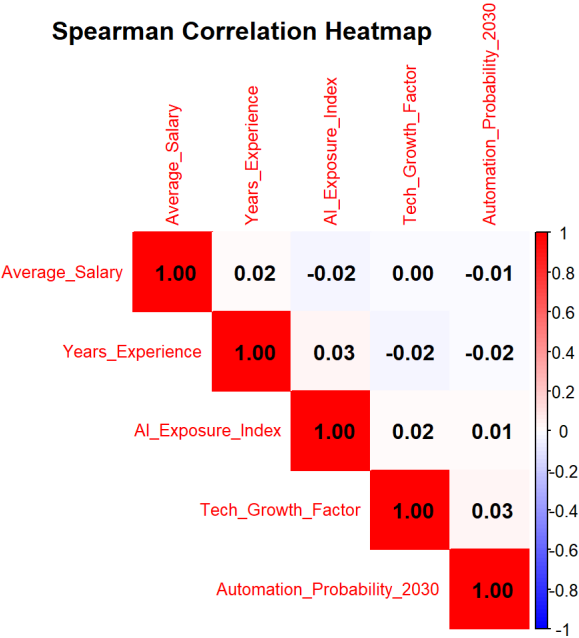




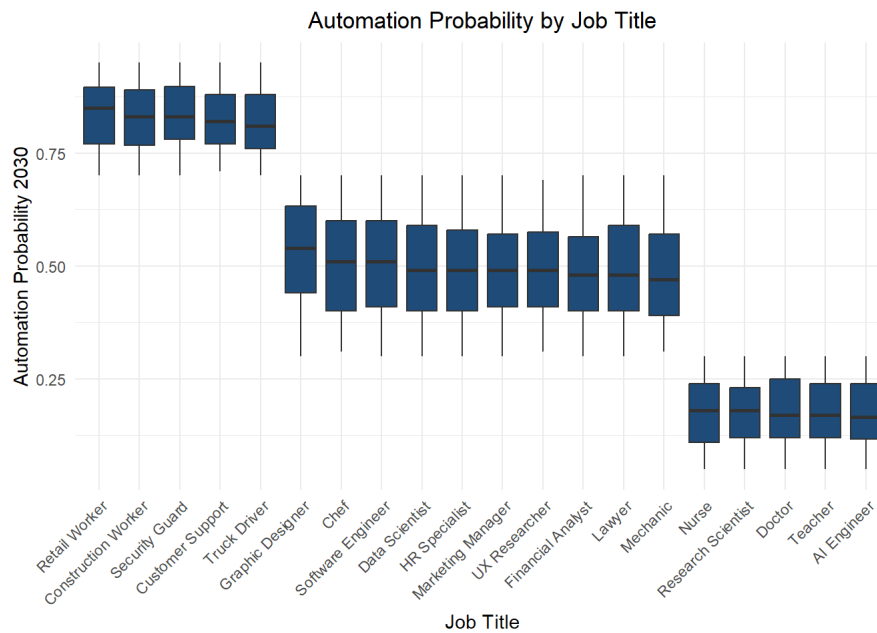
Key Takeaways: Univariate Analysis

- **Average Salary** ranges from \$30,030 to \$149,798, with a median of \$89,318 and mean of \$89,372, indicating a relatively symmetric distribution of salaries across occupations.
- **Years of Experience** spans from 0 to 29 years, with a median of 15 and a mean of 14.7, showing a fairly even distribution with a slight skew towards lower experience levels.
- **AI Exposure Index** varies from 0 to 1, with a median and mean around 0.5, reflecting a balanced distribution of AI dependence across jobs.
- **Tech Growth Factor** ranges from 0.5 to 1.5, with a median of 1.0, suggesting most fields experience moderate technological growth.
- **Automation Probability 2030** spans from 0.05 to 0.95, with a median of 0.5, highlighting that automation risk is distributed across the spectrum, with many jobs at moderate risk.
- **Job Titles** includes 20 unique job titles, with counts ranging from 132 to 175, showing a balanced representation across technical, creative, healthcare, and service occupations.

Multivariate Analysis

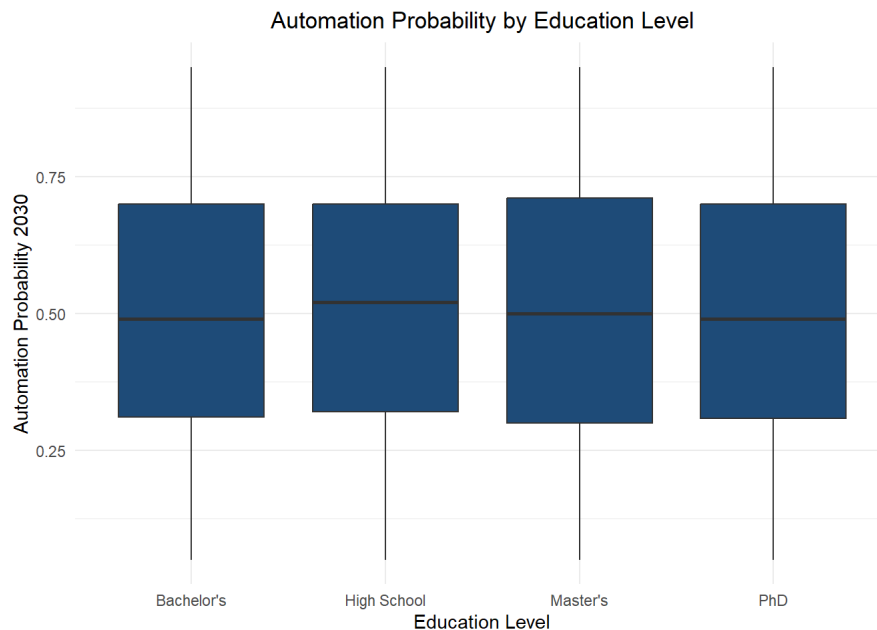


A Spearman correlation analysis was conducted on the key numerical variables (Average_Salary, Years_Experience, AI_Exposure_Index, Tech_Growth_Factor, and Automation_Probability_2030), and the results indicate that the correlations between these variables are very weak, with the coefficients being between 0.03 and -0.02. This suggests that, across all occupations, these numerical features vary largely independently of each other, highlighting the diversity and heterogeneity of jobs in the dataset.

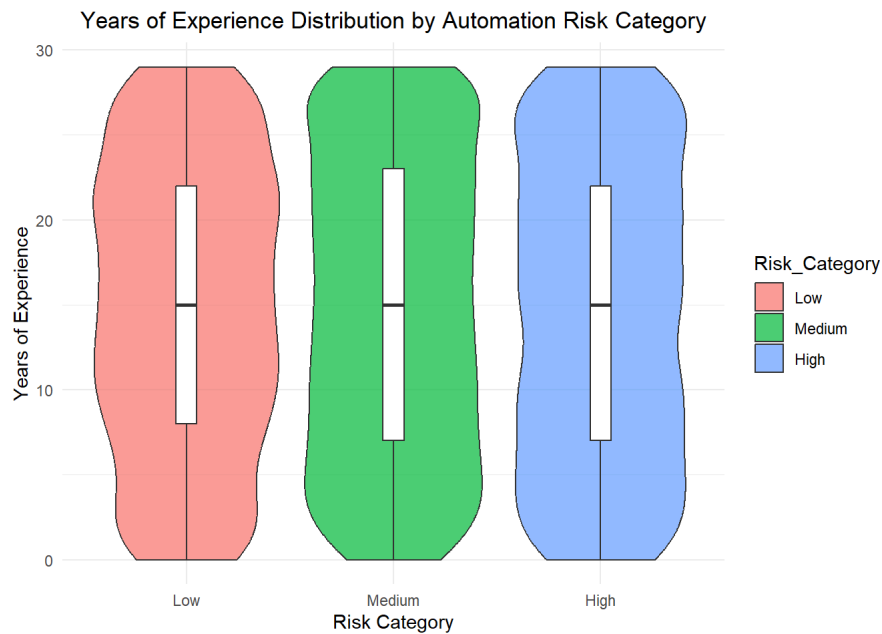
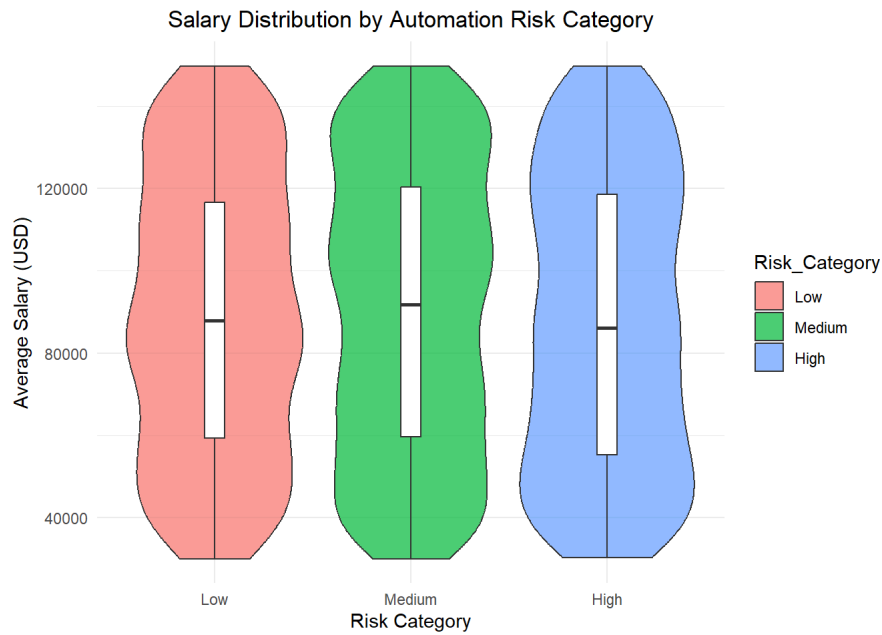


The visualization highlights strong heterogeneity in automation probability across job titles, with routine and manual roles such as Retail Worker, Construction Worker, Security Guard, Customer Support, and Truck Driver exhibiting the highest median automation risk (approximately 0.81–0.85), while more technical, analytical, or creative occupations—including Software Engineer, Data Scientist, Graphic Designer, Chef, and HR Specialist—show lower median probabilities (around 0.49–0.54).

However, the wide dispersion observed within most job titles, with values spanning roughly from 0.30 to 0.95, indicates substantial within-occupation variability. This suggests that job title alone does not fully determine automation vulnerability, reflecting differences in task composition, skill requirements, and exposure to technology even among individuals holding the same role.

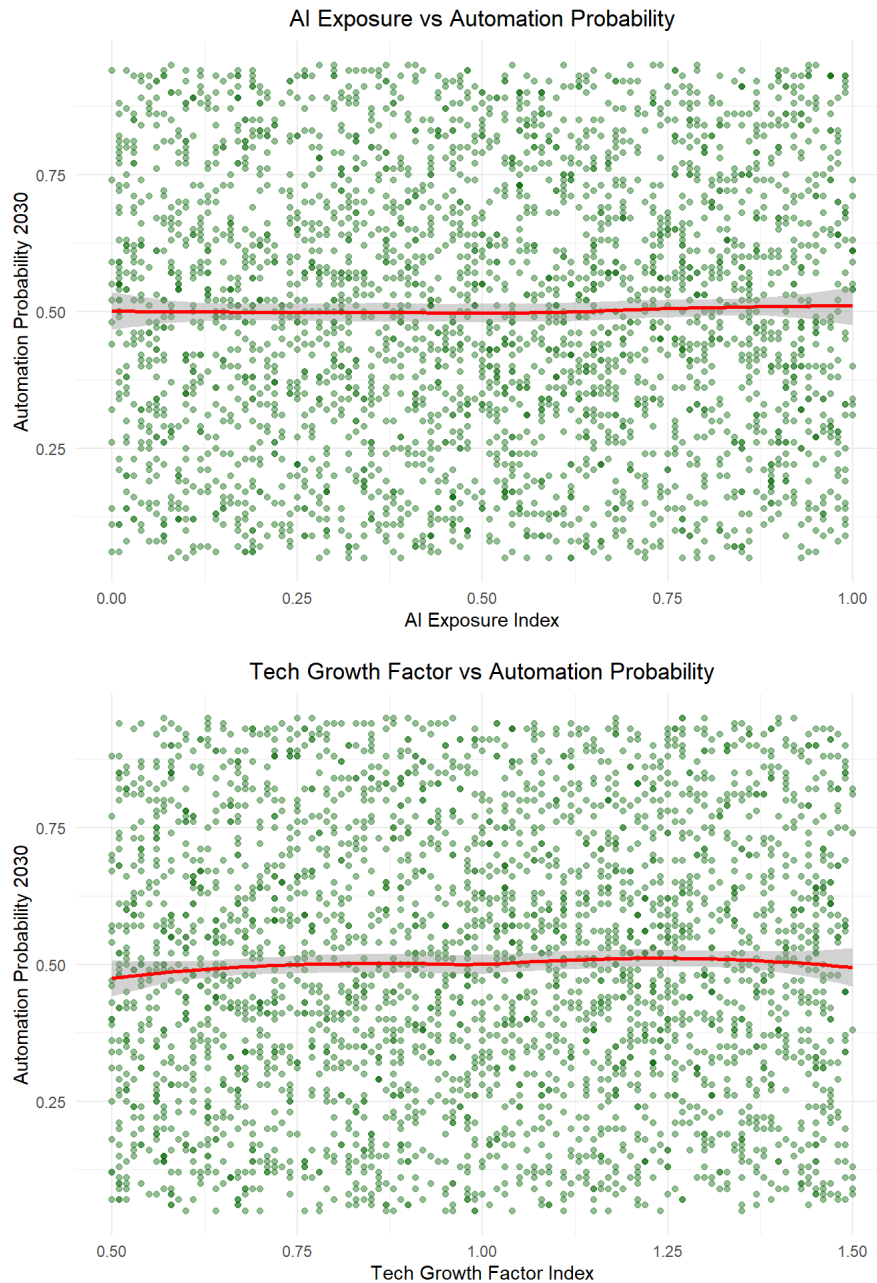


The visualization shows that automation probability varies across education levels, but no clear monotonic relationship is observed. All education categories display substantial within-group dispersion, indicating that jobs requiring similar levels of education can nonetheless face very different degrees of automation risk. This suggests that education level alone is not sufficient to explain automation vulnerability across occupations, and that other factors such as task composition and technological exposure likely play a significant role.



These violin plots illustrate the distributions of salary and years of experience across automation risk categories by combining density estimates with boxplot summaries. For both variables, a strong overlap is observed between low-, medium-, and high-risk jobs, suggesting that salary levels and experience are broadly comparable across risk categories.

While the plots clearly display the spread and shape of each distribution, no automation risk group exhibits a distinctly different pattern. Overall, these results indicate that automation risk is not strongly differentiated by either wage or years of experience in this dataset.



These scatter plots examine the relationships between automation probability and two technological variables (AI exposure and the Tech Growth Factor) using LOESS smoothing to capture potential non-linear patterns. In both cases, the smoothed trends appear relatively flat, and the data points show substantial dispersion across all levels of exposure and technological growth. No strong or consistent association is visually apparent, suggesting that neither higher interaction with AI tools nor faster technological growth within a field is, on its own, a strong determinant of automation risk.

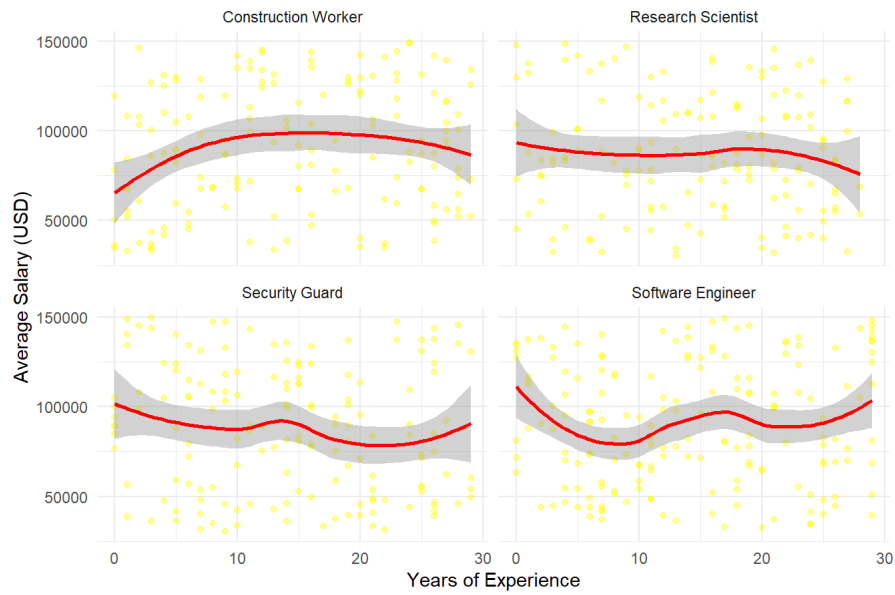
These observations indicate that technological factors alone may have a limited explanatory role in automation vulnerability, and that automation outcomes likely depend on a more complex combination of task characteristics and contextual factors.

Distribution of Automation Risk by Education Level

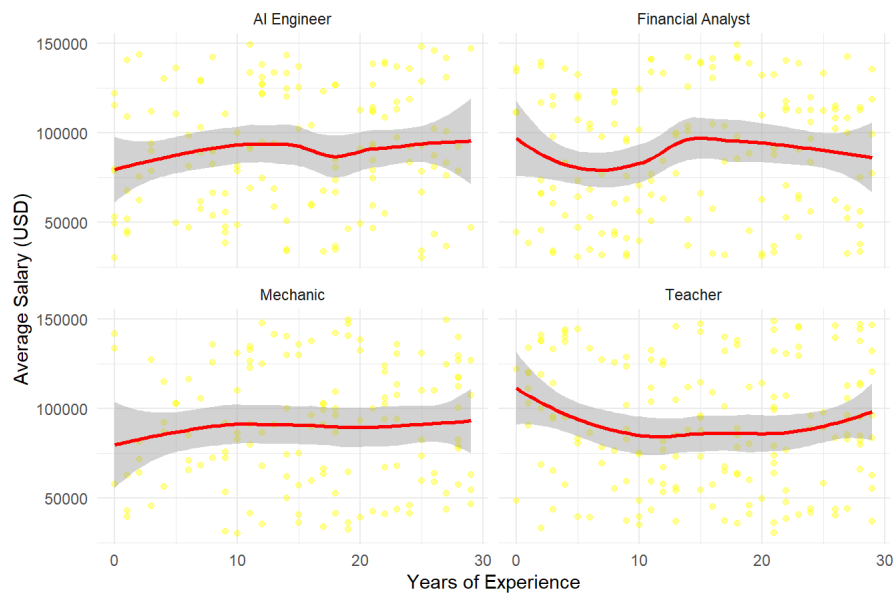
Education_Level	Low	Medium	High
Bachelor's	24.8%	50.6%	24.6%
High School	23.2%	52.9%	23.9%
Master's	25.6%	48.8%	25.6%
PhD	25.0%	50.3%	24.7%

The cross-tabulation shows that automation risk is fairly evenly distributed across all education levels. Most jobs fall into the medium-risk category, while low- and high-risk jobs are roughly balanced for each education level. This indicates that education alone does not strongly determine automation vulnerability, and other factors such as job type, tasks, and technological exposure likely play a more important role.

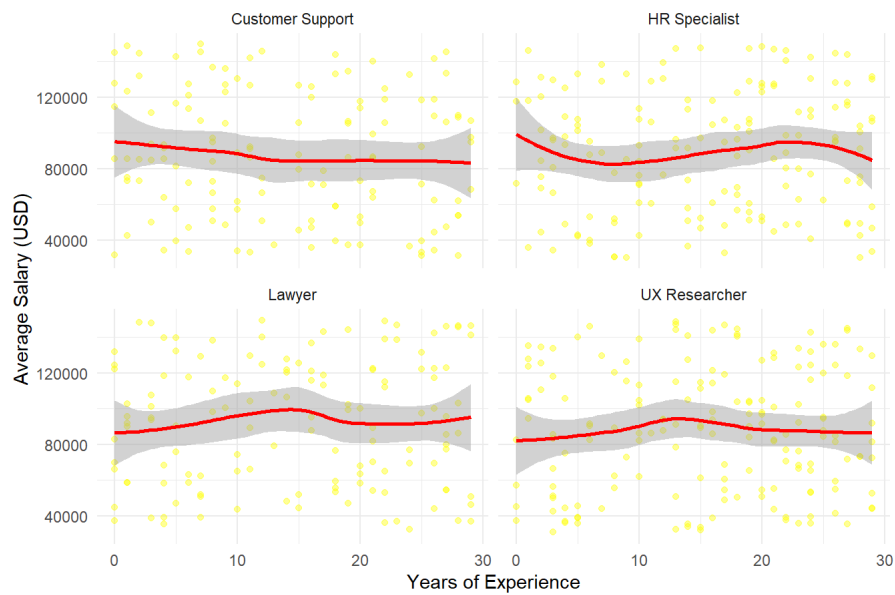
Salary vs Years of Experience by Job Title



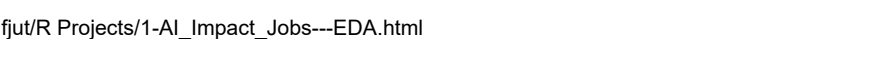
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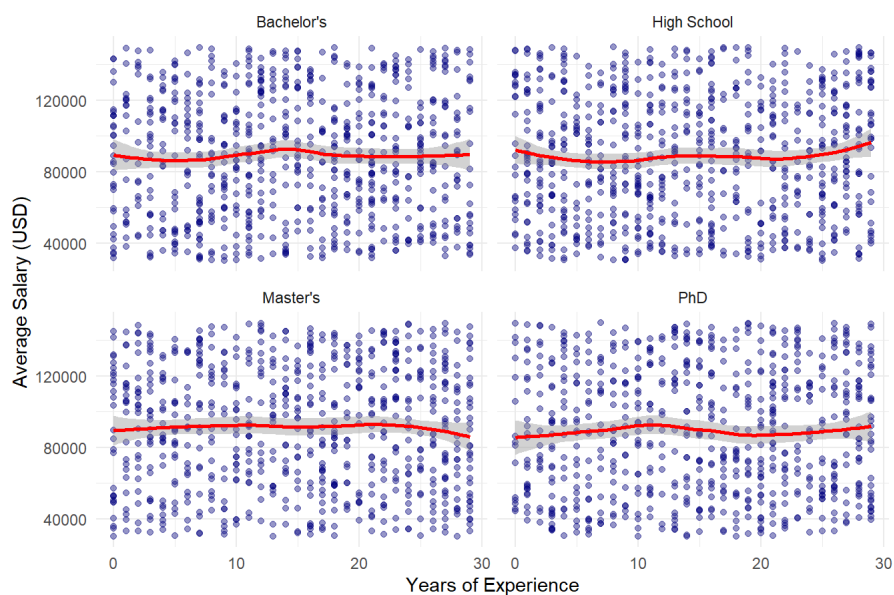
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Salary vs Years of Experience by Job Title

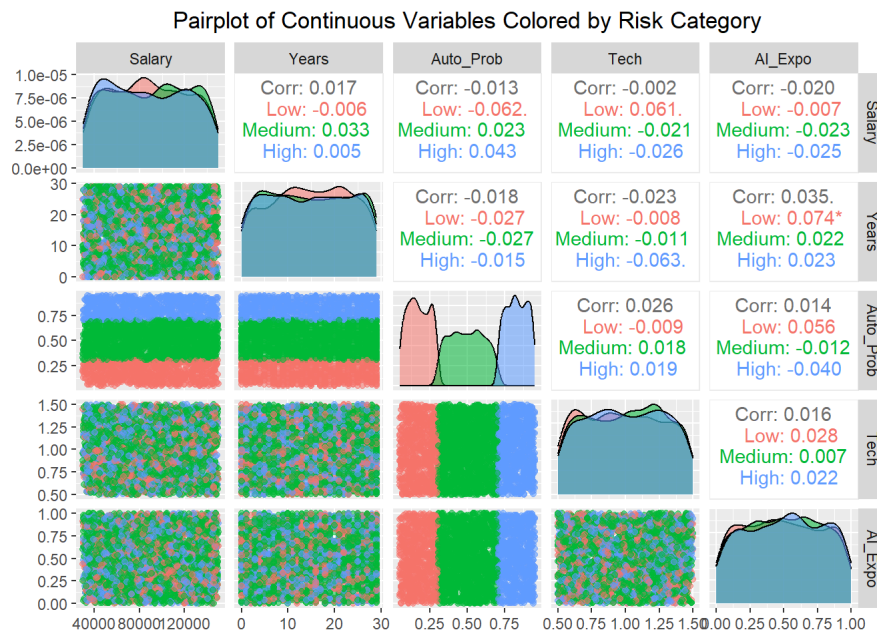
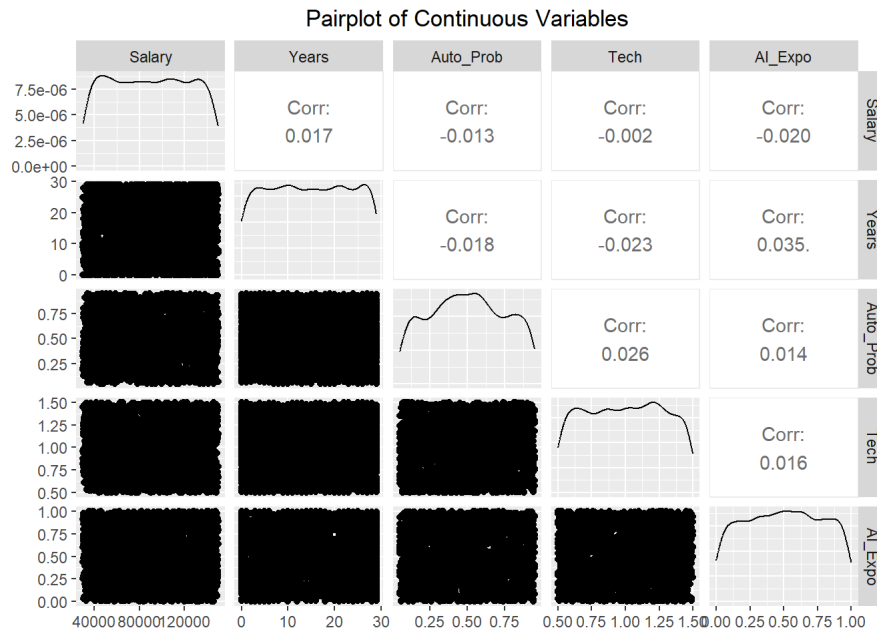


Salary vs Years of Experience by Education Level



These faceted scatter plots describe how salary and years of experience are jointly distributed across job titles and education levels, using LOESS curves to summarize overall patterns. In both cases, the association between experience and salary appears weak and highly variable, with no clear or consistent upward trend as experience increases. Differences in typical salary levels are visible across job titles and education groups, but substantial dispersion is observed within each group.

Overall, these visualizations highlight strong heterogeneity in salary–experience patterns across occupations and education levels, with wide ranges of values present even among individuals with similar experience.



The pairplots, both at the global level and when stratified by automation risk category, reveal very weak associations among salary, years of experience, automation probability, AI exposure, and the tech growth factor. Across all visualizations, the scatterplots are highly dispersed and the corresponding Spearman correlations remain close to zero, indicating the absence of strong monotonic patterns between the continuous variables. This diffuse structure persists within low-, medium-, and high-risk groups, suggesting that stratifying by risk does not uncover more pronounced relationships.

Overall, these visualizations highlight substantial heterogeneity in the data and indicate that the continuous variables vary largely independently across the dataset.

Summary Statistics by Education Level

Education_Level	Mean_Salary	Median_Salary	Mean_Years_Experience	Median_Years_Experience	Mean_AI_Exposure	Median_AI_Exposure	Mean_Tech
Bachelor's	88815.49	89309.0	14.531	14	0.497	0.49	0.49
High School	88389.29	87709.5	14.204	14	0.498	0.50	0.50
Master's	91318.57	92086.0	14.924	15	0.503	0.50	0.50
PhD	89045.57	87818.5	15.101	15	0.508	0.52	0.52

The summary statistics by education level reveal that average salaries are relatively similar across groups, ranging from \$88,389 for High School graduates to \$91,319 for Master’s degree holders, with median salaries showing the same modest differences.

Years of experience are also fairly consistent, with mean values between 14.2 and 15.1 years, indicating that higher education levels do not correspond to substantially more experience on average.

AI exposure and Tech Growth factors are balanced across education levels, with mean AI exposure around 0.50 and Tech Growth factors close to 1, suggesting that the technological environment does not differ dramatically by education.

Finally, automation probability is broadly similar across groups, with median values near 0.49–0.52, highlighting that education alone does not strongly differentiate occupations in terms of automation risk.

Overall, these results indicate modest differences across education levels, and suggest that other factors beyond education likely drive variation in salary, AI exposure, and automation probability.

Summary Statistics by Job Title

Job_Title	Mean_Salary	Median_Salary	Mean_Years_Experience	Median_Years_Experience	Mean_AI_Exposure	Median_AI_Exposure	Mean_Technical_Growth
AI Engineer	90759.83	90705.0	14.443	14.0	0.474	0.445	1.000
Chef	91407.80	94043.0	14.929	15.0	0.529	0.520	1.000
Construction Worker	90800.33	89613.0	14.558	14.5	0.505	0.550	1.000
Customer Support	87487.00	86153.0	14.000	13.0	0.502	0.530	1.000
Data Scientist	90629.55	91063.0	13.898	13.0	0.513	0.510	1.000
Doctor	89091.74	88326.5	14.674	14.0	0.505	0.490	1.000
Financial Analyst	88030.31	91146.0	14.351	14.0	0.464	0.470	1.000
Graphic Designer	96423.47	103183.0	14.631	13.0	0.527	0.510	1.000
HR Specialist	89242.26	93115.0	15.439	17.0	0.489	0.500	1.000
Lawyer	92441.88	95140.0	14.720	15.5	0.509	0.495	1.000
Marketing Manager	86458.11	85592.0	13.515	12.0	0.524	0.520	1.000
Mechanic	89807.65	91075.5	16.228	17.0	0.457	0.445	1.000
Nurse	88955.32	91926.0	14.000	14.0	0.521	0.540	1.000
Research Scientist	87204.74	84544.0	14.839	16.0	0.479	0.480	1.000
Retail Worker	87208.41	84150.0	14.839	16.0	0.506	0.520	1.000
Security Guard	87290.43	88584.5	14.201	14.0	0.515	0.525	1.000
Software Engineer	90264.74	87803.0	14.177	14.0	0.513	0.540	1.000
Teacher	90885.76	88668.0	15.524	16.0	0.497	0.475	1.000
Truck Driver	84519.93	81290.0	14.876	15.0	0.520	0.510	1.000
UX Researcher	87982.42	87737.0	15.581	16.0	0.476	0.490	1.000

The summary statistics by job title reveal noticeable differences in average automation probabilities, salaries, and other job characteristics across occupations. For instance, roles such as Retail Worker, Construction Worker, Customer Support, and Security Guard tend to have higher median automation probabilities (around 0.82–0.85), while technical and professional positions like AI Engineer, Doctor, Nurse, Research Scientist, and Teacher show lower median probabilities (around 0.17–0.18).

Salaries vary across job titles, with higher averages observed for positions such as Graphic Designer, Lawyer, and Chef, and lower averages for roles like Truck Driver, Marketing Manager, and Retail Worker.

Years of experience, AI exposure, and Tech Growth factors appear fairly consistent across most occupations, with moderate variation between jobs.

Overall, these patterns suggest that automation risk, compensation, and technological exposure differ by job title, but job title alone does not fully explain these differences, highlighting the heterogeneity of occupations in the dataset.

Key Takeaways: Multivariate Analysis

- **Automation risk varies widely across occupations:** Jobs in the dataset span the full range of automation probabilities, with a large share of roles concentrated around moderate levels of risk rather than at extreme values.
- **No strong associations emerge between automation risk and common job characteristics:** Salary, years of experience, and education level display very weak correlations with automation probability, indicating that these variables do not show strong monotonic relationships with automation risk in this dataset.
- **Education level alone does not clearly differentiate automation risk:** Considerable dispersion in automation probability is observed within each education category, suggesting that jobs with similar educational requirements can be associated with very different levels of automation exposure.
- **Higher AI exposure is not systematically aligned with higher automation probability:** Across the data, jobs with greater interaction with AI tools do not consistently exhibit higher automation risk, pointing to heterogeneous patterns rather than a clear association.
- **Automation risk spans the wage distribution:** Salary distributions overlap substantially across low-, medium-, and high-risk categories, indicating that automation risk is present across different wage levels rather than being concentrated at one end of the income spectrum.
- **Overall patterns highlight strong heterogeneity:** Taken together, the multivariate descriptive results suggest that automation risk is distributed unevenly across jobs and is not strongly structured by any single observed characteristic, reflecting the complex and diverse nature of occupations in the dataset.